

Hand-in set 6 in FYS2130, spring 2026

General instructions. For all problems, you must clearly explain your reasoning and explicitly state any assumptions you make. Answers without justification will receive little or no credit. Where sketches or qualitative arguments are requested, they must be included and clearly labeled. Use the methods presented in the lectures.

Problem 1. Two identical loudspeakers emit monochromatic sound of frequency $f = 440$ Hz. The speed of sound is $c = 340$ m/s. The speakers are driven in phase and emit waves of equal amplitude A_0 .

The speakers are placed along the x -axis at $x = 0$ and $x = d$, with $d = 1.2$ m. The observation direction is in the far field and specified by the angle θ measured from the y -axis.

Although the sources are in phase, the waves generally arrive out of phase due to different path lengths, producing an interference pattern.

1. Let r_1 be the distance from the observer to the source at the origin, and r_2 the distance to the source at $x = d$. Show that, in the far-field approximation $r \gg d$,

$$r_2 \approx r_1 - d \sin \theta,$$

and hence that the path difference is

$$\Delta r = |r_2 - r_1| = d \sin \theta.$$

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Hint: Use the Taylor approximation $\sqrt{1+a} \approx 1 + a/2$.

2. Show that this leads to a phase difference $\Delta\phi$. By superposing the two waves in complex form, express the result as

$$y(\theta, t) = A(\theta) e^{-i\omega t},$$

and show that

$$A(\theta) = 2A_0 \cos\left(\frac{\Delta\phi}{2}\right) = 2A_0 \cos\left(\frac{kd \sin \theta}{2}\right).$$

You may find it helpful that

$$1 + e^{ia} = 2 \cos\left(\frac{a}{2}\right) e^{ia/2},$$

3. Determine the wavelength λ from the given frequency.
4. Show that the sound intensity is

$$I(\theta) = I_0 \cos^2\left(\frac{kd \sin \theta}{2}\right).$$

Problem 2. A beam of electrons is emitted with kinetic energy $E = 150 \text{ eV}$. The electrons pass through a double-slit apparatus and are detected on a distant screen.

The two slits act as coherent sources separated by a distance $d = 2.0 \mu\text{m}$. We assume the far-field regime, so that each detection point on the screen is characterized by an angle θ measured from the normal to the slit plane.

At a point on the screen, the quantum wavefunction is approximated as the coherent sum of two contributions:

$$\psi(\theta) = \psi_0 \left(e^{ikr_1} + e^{ikr_2} \right),$$

where the phase difference arises from the path difference

$$\Delta r = d \sin \theta.$$

The electron momentum and wave number are related to the energy by

$$p = \sqrt{2mE}, \quad k = \frac{p}{\hbar}.$$

1. Compute the de Broglie wavelength λ and wave number k of the electrons.

Using the phase difference $k\Delta r$, rewrite the wavefunction in the form

$$\psi(\theta) = A(\theta) e^{i\phi(\theta)},$$

and show that the probability density can be expressed as

$$P(\theta) \propto \cos^2 \left(\frac{kd \sin \theta}{2} \right).$$

You may find it helpful that

$$\cos a = \frac{1}{2}(e^{ia} + e^{-ia})$$

2. Derive the condition for interference maxima and minima in terms of θ .

In the small-angle approximation $\sin \theta \approx \theta$, determine the angular spacing between adjacent maxima.

3. Plot $P(\theta)$ for $\theta \in [-5^\circ, 5^\circ]$.

Briefly describe how this pattern differs from classical particle trajectories, and explain why the same interference pattern emerges even when electrons are emitted one at a time.